

Question	Answer	Marks
5(a)	potential is (defined as) zero at infinity	B1
	proton has a positive charge and so repels another positive charge	B1
	work is done on two (positive) charges to move them towards each other or work is done by two (positive) charges as they move apart from each other	B1
5(b)(i)	arrow drawn through Q in a WSW direction directly away from the centre of the sphere	B1
5(b)(ii)	curve in at least one quadrant passing through $(\pm R, \pm E_0)$ and $(\pm 2R, \pm \frac{1}{4}E_0)$	B1
	curve between $-3R$ and $-R$ of increasing magnitude of gradient and curve between R and $3R$ of decreasing magnitude of gradient	B1
	two lines drawn, one in the top right quadrant, the other in the bottom left quadrant	B1
5(c)	$E_P = - (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (5.3 \times 10^{-11})]$	C1
	$= -4.3 \times 10^{-18} \text{ J}$	A1

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